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A new Devonian palaeomagnetic pole for the Russian platform and Baltica, and related apparent polar wander

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SUMMARY

Comparison of published palaeomagnetic data from the Russian platform and Britain for Mid-Palaeozoic times places an ocean between the two of the order of 2000 km latitudinal width. This contradicts geological evidence, which favours contiguity between the two continental units by Devonian times. We propose that this discrepancy can be attributed to the incorrect dating of palaeomagnetic remanence. We strove to remedy this situation by conducting a new palaeomagnetic investigation of Silurian and Devonian sediments in west Ukraine, USSR (latitude 48.7°N, longitude 26°E). That investigation uncovered a pervasive Upper Palaeozoic remagnetization corresponding to a palaeomagnetic pole at lat. 45.8°S, long. 348.2°E, dp/dm 1.6°/3.1°, $N = 25$ sites (reversed polarity). A second primary or near-primary Devonian remanence was identified with pole at lat. 3.7°N, long. 325.5°E, dp/dm 5.2°/8.7°, $N = 25$ samples (mixed polarity). The age determination for this Devonian result is based on its dual polarity, evidence for a detrital remanent mode, and early host mineral phase (haematite). The new Devonian result for the Russian platform is conformable with the Devonian parts of the British apparent polar wander paths and therefore eliminates the aforementioned discrepancy between palaeomagnetic and geological evidence for the timing of assembly of the 'Old Red Continent'.

Key words: Baltica, detrital remanent magnetization, Devonian, palaeomagnetism, plate reconstruction, Russian platform.

INTRODUCTION

There is a longstanding palaeomagnetic discrepancy between Devonian rocks of the Russian platform and Britain, contradicting conventional geological wisdom which states that the two regions had their present relative configuration as parts of the 'Old Red Continent' by that time. Mid-Palaeozoic palaeogeographic reconstructions based on palaeomagnetic data from the Russian platform and Britain feature a Medio-European ocean of the order of 2000 km palaeolatitudinal width. It is clear that the discrepancy either reflects the incorrect dating of palaeomagnetic directions or that what we think of as the 'Old Red Continent' was not assembled until after the onset of Devonian Old Red Sandstone facies sedimentation.

The most likely explanation for the aforementioned discrepancy is expected to lie in the dating of palaeomagnetic directions carried by sediments of the Russian platform: it is notoriously difficult to reliably date the palaeomagnetic remanence carried by flat-lying sediments. If a wrong age is attributed to one of the two data sets in the comparison, then the comparison is, of course, invalid.

It is notable that most of the reported palaeomagnetic poles from the Russian platform of Devonian, Carboniferous and Permian age are approximately coincident. Furthermore, they correspond in position to the Upper Palaeozoic portion of the British apparent polar wander (APW) path. Therefore there is circumstantial evidence for differential remagnetization of rocks of the 'Old Red Continent'; for remagnetization of Mid-Palaeozoic rocks of the Russian platform in Upper Palaeozoic time, evidenced by a temporary cessation in APW. If much of the data from Russian Mid-Palaeozoic rocks are indeed of Upper Palaeozoic (Kiaman?) age, then the major palaeomagnetic discrepancy between the Russian platform and Britain is

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Table 1. Selected palaeomagnetic pole positions for the Russian plate.

Pole N°	Rock unit	Rel.	Study Lat. (°N)	Study Long. (°E)	Remanence Dec. (°)	Remanence Inc. (°)	α_{95} (°)	P	Age Ma	Pole Lat. (°N)	Pole Long. (°E)
Permian											
07062	Red siltstones, river Volga	A	56	44	44	43	3	n	249	45	159
57	Sediments of north Dvina	B	60.8	46.5	224	-41	4	m	249	-41	348
60	Red clay, river Viatka	B	59	50.8	225	-47	4	m	249	-47	347
59	Red clay, river Soukhona	B	60.6	44.2	221	-45	3	m	250	-46	346
61	Red clay, river Volga	B	55	49	221	-44	6	m	250	-48	347
06	Red clay, south Pre-Ural	C	52.5	55	218	-48	3	m	250	-53	350
64	Multicoloured rocks, Volga	A	55	49	226	-42	5	m	250	-45	342
58	Red clay, river Soukhona	B	60.8	45.2	223	-45	6	m	251	-45	346
63	Red clay, river Volga	B	56	49	222	-42	12	r	251	-45	347
84	Red clay/marl/sst Povolzche	A	55	49	224	-44	2	m	251	-47	343
24	Red clay, river Viatka	B	56	51	222	-37	4	r	252	-43	352
79	Red clay, sltst, sst, r.Kama	B	56.6	54	229	-41	5	r	253	-41	346
80	Sediments, Prikame	A	54.9	53	221	-41	8	r	254	-46	352
25	Red clay, m. seds. r.Viatka	B	56	51	225	-39	6	r	255	-43	348
67	Sediments, river Kama	B	56	51.7	222	-43	4	r	255	-47	350
69	Sediments, river Prikame	B	54.6	52.6	224	-44	2	r	256	-46	347
71	Sediments, northeast Tataria	B	55.5	52.7	223	-40	3	r	256	-45	351
70	Sediments, Prikame	B	56.2	56.5	235	-41	3	r	257	-37	344
91	Arg/sltst/sst/lst, S. Urals	A	50.4	58	228	-34	5	r	275	-40	350
21	Red clay, Donbass	B	48	38	218	-20	5	r	284	-41	345
20	As #21	B_1	48.7	38.2	223	-18	5	r	285	-37	342
92	As #91	B^T	50.7	58	236	-15	7	r	285	-27	351
Carboniferous											
08002	Red clay/sltst, Donbass	B	48.4	38.2	208	-22	4	r	286	-46	356
01	Red clay, Podmoskovye	A	55.5	38.5	215	-31	5	r	290	-43	350
05	Red clay, Rzhev	C	56.5	34.5	225	-21	4	r	298	-33	338
08	Red clay, Ozery	C	55	38.5	220	-15	7	r	300	-33	349
07	Red clay, Serpoukhov	C_1	55	37.5	232	-18	5	r	300	-29	335
35	Arg and sltst, Donbass	C^T	48	37	225	-36	12	r	300	-44	330
51	Lst, sltst, sst, Donbass	B^T	48	38	226	3	8	r	300	-26	344
09	Red clay, Shatsk	C	55	42	224	-13	6	r	305	-31	348
06	Red clay, Vereya	C_1	55.5	36	221	-14	7	r	310	-32	346
52	Lst, sltst, sst, Donbass	B^T	46	38	219	11	8	m	315	-26	354
14	Red clay,Tikhvin/Bauxitogorsk	B	59.4	34	218	-42	6	m	333	-46	340
72	Limestone, south Urals	A_1	53	57	265	-14	10	m	356	-8	326
91	Sandstone, south Urals	B^T	55	58.5	223	-46	6	r	358	-43	339
Devonian											
09083_1	Ophiolites, south Urals	A^T	48	59	275	-29	11	m	384	-8	315
83_2	As above	A^T	48	59	281	-30	5	m	384	-5	310
83_3	As above	B^T	48	59	102	30	6	n	384	4	129
Silurian											
10005	Red sltst, Zmeinyi, Black Sea	B	45	30	261	2	6	r	428	-6	307
Ordovician											
11036	Kourganskyi suite, S. Urals	C_1	51.5	57.8	338	-33	6	m?	455	19	260
22	Red seds Kouragancku su. S.U.	B^T	51.5	58	141	29	14	n?	478	-14	96
42	Sills and baked seds, S.Urals	B_1	51	58	152	36	24	n?	485	-15	78
27	Sst, sltst, south Urals	B^T	51.2	58.1	175	15	6	n?	488	-30	64
Silurian and Devonian Data not used in the calculation of the APW Path											
09084	Lst, sst, Zilairsky syncline	C_1	53	57	257	23	4	m	386	2	341
10026	Sediments, river Belaya	C^T	53	57	250	25	3	m	428	-1	251
10027	Volcanics and seds, S. Urals	B^T	51	57.5	227	30	4	m	425	-11	012

Explanation for Table 1.

Pole N°: according to the lists of Khramov (1984, 1986).

Rel: palaeomagnetic reliability based on the scheme of Briden & Duff (1981; also see ^{T} below).

^{T}: Tectonic rotation about a vertical axis, on a scale larger than the study area, cannot be discounted.

P: polarity (r)eversed, (m)ixed or (n)ormal. Note that remanence directions and pole positions are given in the polarity as indicated under 'P'. Directions and positions for (m)ixed polarity data are given in reversed polarity. '?' indicates data with uncertain/unknown polarity.

Abbreviations: Arg=argillite, sst=sandstone, lst=limestone, sltst=siltstone, seds=sediments, m=marine, su=suite, S=south, U=Urals.

Note that the palaeomagnetic pole positions listed in Table 1 are illustrated in *reversed polarity* form in Figs 2 and 3. Two polarity possibilities are given for the Ordovician results, one in Fig. 2 and the other in Fig. 3.

eliminated, yet we are left with a large gap in the record of Palaeozoic APW for the Russian platform. It is perhaps significant that results reported to be of Devonian age from two rock units in the USSR (Table 1) resemble Devonian results from Britain. They are, however, of limited reliability and should be interpreted with caution.

It is timely that a re-evaluation of palaeomagnetic data from the Russian plate (including the Russian platform) be made, enabling proper comparison of Russian and British data. We have already intimated that this would leave a large Mid-Palaeozoic gap in the APW record for the Russian plate. Therefore new field work and palaeomagnetic analysis have been carried out on Siluro-Devonian rocks of the Russian platform in an attempt to unveil demonstrably primary remanent magnetizations and so fill in the gap in APW. This paper offers a detailed description of those new data and of their implications for the timing of assembly of the 'Old Red Continent'.

REFERENCE PALAEOMAGNETIC DATA SETS

We compare the new data presented in this paper with the APW paths for Britain given in Torsvik *et al.* (1990; Fig. 1), based on the data selection of Briden *et al.* (1987) as updated by Torsvik *et al.* (1990). We also generate a new APW model for the Russian plate and test it against our new data.

OUR RESELECTION OF DATA FROM THE RUSSIAN PLATE

The close correspondence between the majority of published palaeomagnetic data from the Russian plate, from Siluro-Devonian sedimentary lithologies, and the Upper Palaeozoic section of the British APW path was noted in the introduction and is illustrated in Fig. 1. We attribute this discrepancy and the consequent apparent Mid-Upper Palaeozoic cessation in APW to widespread Permo-Triassic remagnetization of Mid- and Lower Palaeozoic rocks: sediments in particular. For this reason we did not consider palaeomagnetic data from Mid-Palaeozoic sediments which fell within the group of poles illustrated in Fig. 1.

Palaeomagnetic pole positions from rocks of Ordovician to Permian age from the Russian plate (Khramov 1984, 1986) have been classified according to the reliability criteria of Briden & Duff (1981; 'A' to 'D': most to least reliable). In our present assessment of palaeomagnetic reliability we consider the possibility of large-scale rotation of blocks along the former continental margins outside their scheme. If the rotation of whole study areas about a vertical axis cannot be discounted, pole grades are amended to 'A^T' to 'D^T'. Those data which we deemed to be the most significant are listed along with their reliability categories in Table 1.

The selected data are plotted in Fig. 2 as palaeomagnetic south poles, assuming their reported polarities to be

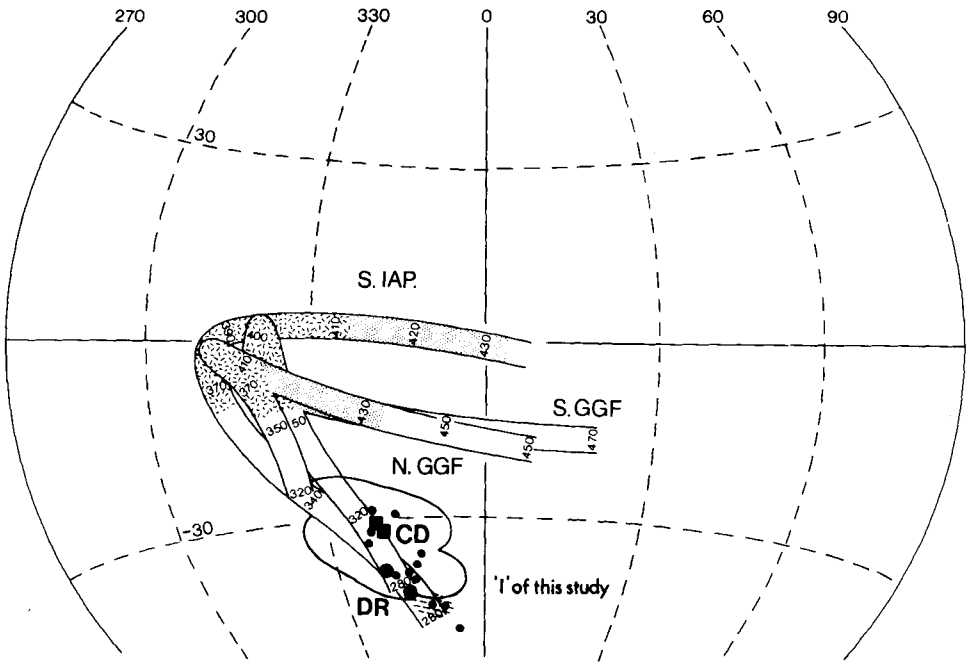


Figure 1. Reference south APW paths for Britain from Torsvik *et al.* (1990) based on the data selection of Briden *et al.* (1987) as updated by Torsvik op. cit. Small dashes = Devonian, stipple = Silurian, ages are in Myr. S.IAP = south of Iapetus suture, S.GGF = south of the Great Glen Fault, N.GGF = north of the GGF. DR (large circles) = mean poles from the Siluro-Devonian Dniester River section, Podolia (Khramov *et al.* 1981; Hicken *et al.* 1972). CD (large squares) = combined 'Devonian' poles for the Russian platform (Main Devonian Field, Hicken *et al.* 1972). Small circles = palaeomagnetic poles from Devonian sediments of the Russian platform. A curved 'error envelope' encloses Russian pole positions for which 95 per cent error estimates are available. Pole with shaded error oval = 'I' component of this study (adjacent to 280 Myr label).

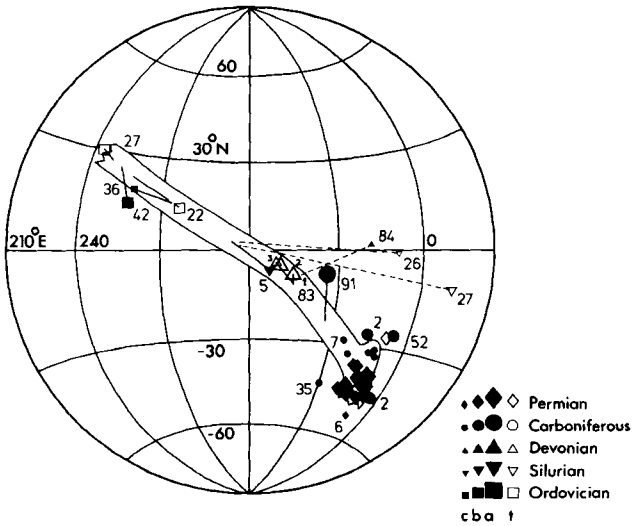


Figure 2. Selected south palaeomagnetic poles for the Russian plate (Table 1). Note that where practical the last two digits of the pole number in Table 1 is written beside palaeomagnetic poles. 'a', 'b' and 'c' are reliability categories according to Briden & Duff (1981; 'a' is the most reliable). Open symbols ('t' in legend) indicate those data for which large-scale tectonic rotations have not been ruled out. A smooth (APW) path fitted to the data using the algorithm of Jupp & Kent (1987) is indicated. Note that where possible palaeopole positions are linked with straight lines to the parts of the fitted path to which they correspond in time. Data linked to the path with dashed lines were not used in the derivation of the path. Longitude–latitude graticules are in 30° intervals. Projection: Lambert equal area.

reliable, along with a smooth path fitted to the data using the 'spherical spline' algorithm of Jupp & Kent (1987). Inevitably, following the aforementioned exclusion of data from some Mid-Palaeozoic rocks, the Mid-Palaeozoic part of our new path is largely interpolated, leading to

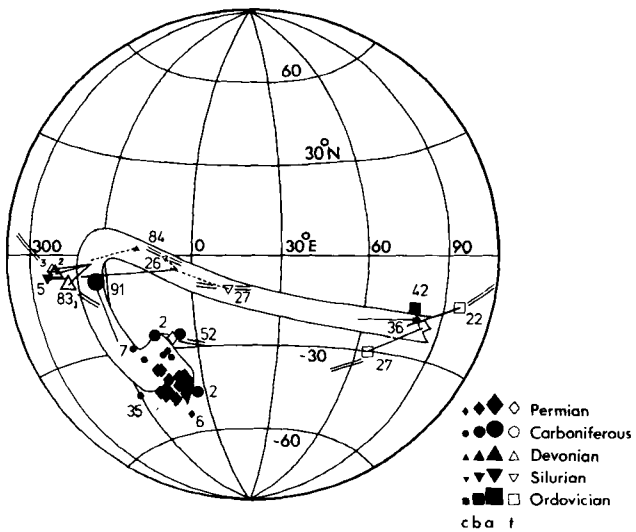


Figure 3. An alternative APW path to that given in Fig. 2, with the polarity of the Ordovician data inverted. Double lines have been added beside those poles for which tectonic rotation of the whole study area, about a vertical axis, cannot be discounted. The double lines indicate the directions in which the pole would move if correction for such a rotation were made.

uncertainty in the polarity of the Ordovician data. Therefore our selection of data is plotted in Fig. 3 with the polarity of the Ordovician data inverted (i.e. with the Ordovician data translated through 180° from their positions in Fig. 2). The second APW path constitutes our favoured model. This is because (1) that path possesses, as it should, the characteristic Mid-Palaeozoic 'corner' defined by southern British data (Trench & Torsvik 1991), (2) the path passes close to three Mid-Palaeozoic data from the Russian plate which must otherwise be considered anomalous, and (3) the path is in closer agreement with that for Scandinavia [see the assessment of Scandinavian versus Russian data offered in Smethurst (1991)]. As far as we are aware, the original polarity interpretation for the Ordovician data from the Russian plate was influenced by the polarity of the better defined Ordovician data from Siberia (Khramov, Petrova & Pechersky 1981). In other words, the APW interpretation given in Fig. 2 has previously been used because it minimizes the post-Ordovician relative rotation required between the Russian and Siberian plates. To the contrary, we feel that the Ordovician data for Siberia offer no constraints on the polarity of the Russian data and therefore endorse the second interpretation (Fig. 3). We must, however, express some reservations regarding the exact locus of the Russian path through Ordovician time in the light of evidently very reliable data from Ordovician limestones in Sweden (Torsvik & Trench 1991a, b). Those Scandinavian data imply that the Ordovician part of the south APW path for Baltica lies north of our 'Russian path'. This controversy, described fully in Smethurst (1991), does not impinge greatly on the subject of this paper.

Note at this point that previously published palaeomagnetic data for the Dniester River sequence (Fig. 1), sampled as part of this study, fall on the Permian part of both the reference British APW paths, and our new APW path for the Russian plate.

THE CHOICE OF PALAEOMAGNETIC SAMPLING REGION

Silurian limestones and Devonian red sand and siltstones outcropping along the Dniester River in Podolia, west Ukraine (Fig. 4), were selected as promising rocks for palaeomagnetic analysis; with the specific intention of resolving the aforementioned palaeomagnetic discrepancy. The previous palaeomagnetic analyses of these rocks had indicated multicomponent natural remanent magnetization (NRM) but had furnished few constraints on the direction of an elusive high-stability remanence component. Moreover, those rocks are (1) well-dated biostratigraphically, (2) of an appropriate age, (3) varied in lithology, and (4) accessible.

In this study a palaeomagnetic sampling site consists of an area of outcropping rock a few metres across from which about eight drilled cores and three or four block samples were taken. 28 such sites were sampled in a traverse through the Silurian and Devonian stratigraphy between Ataky and Dolian (Fig. 4). Representatives of most of the well-exposed sedimentary lithologies in that sequence were sampled.

LITHOLOGICAL DISTRIBUTION

The flat-lying Palaeozoic sediments outcropping in the Dniester River area have both marine and terrestrial

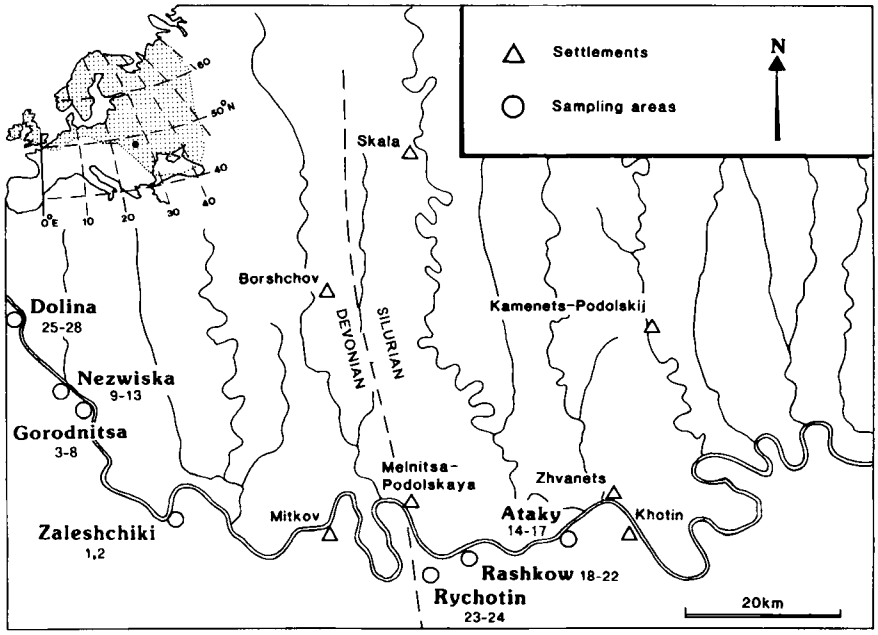


Figure 4. Palaeomagnetic sampling areas. The inset, top left, shows the location of Podolia and the sampled Dniester River stratigraphic section. Site numbers are indicated against sampling area names. The Devonian (resp. Silurian) rocks are confined to the western (resp. eastern) part of the map. The stratigraphic order of sampling sites is indicated in Table 2.

origins. The Silurian succession is essentially shallow marine, consisting of *c.* 500–650 m of limestones, marls, shales and dolomites. More specifically, the sediments at Rashkow and Rykhotin consist of grey marls, dark bituminous limestones, grey bedded limestones and shales. At Ataky granular limestones, yellow flaggy dolomites and grey compact limestones predominate.

Most of the Devonian succession is of continental origin. It consists of between 1000 and 1800 m of red and buff sandstones, siltstones and limestones. In detail, at Zaleshchiki the sequence passes upwards from marine limestones, grey shales and sandstones of earliest Devonian/latest Silurian age into continental red shales and mudstones. At Gorodnitsa and Nezwiska the Lower Devonian sediments consist of well-bedded red and buff sandstones and shales. Higher in the sequence at Dolina the Devonian rocks consist of red siltstones and sandstones along with massive units of coarse grey sandstone.

Detailed sedimentary logs recording the Mid-Palaeozoic successions of Podolia can be found in the 'Guide Book for the Geological Excursion on Silurian and Lower Devonian Sediments of Podolia' (1968). A relatively recent account of the stratigraphy of the area, used in the construction of Tables 2–4, is included in the 'Transactions of the Geological Congress, 27th Session, Combined Guide Book' (tour 095; 1984). The reader is also referred to Nalivkin (1962), Khain (1985), Dineley (1979) and Heckel & Witzke (1979) for more geological information.

PALAEOMAGNETIC ANALYSIS

The palaeomagnetic remanence components reported here are derived from complete progressive thermal and alternating field (AF) demagnetization experiments. Remanence measurements were performed using a CCL cryogenic magnetometer; linear and planar segments in demagnetization trajectories were identified and tested using the LINEFIND algorithm of Kent, Briden & Mardia (1983). We should also point out at this early stage that we define the polarity of our new data in accordance with the Silurian and younger APW interpretation offered in Fig. 3. Characteristic demagnetization trajectories are given in Fig. 5 (Silurian limestone), Fig. 6 (Devonian redbeds), Fig. 7 (Devonian siltstones) and Fig. 11(a) (Silurian limestone AF demagnetization). Unblocking temperature (T_b) spectra corresponding to individual remanence components are given in Figs 5–7. Note that the Silurian and Devonian sediments in Podolia are flat lying and therefore stratigraphic correction of remanence directions is inappropriate.

REMANENCE COMPONENTS

Limestone

Silurian and Devonian limestone samples (sites 14 to 24, and 1 respectively) were found to carry two components of NRM. Example demagnetization trajectories in Figs 5 and 11(a) demonstrate that those two components have essentially separate (L)ow and (I)ntermediate T_b and coercivity (H_{cr}) spectra ('L': <200 °C, <7.5 mT, 10–60 per cent of total remanence, Table 2; 'I': 200°–510 °C, 7–70 mT, Table 3). Furthermore, the direction of components 'L' and 'I' change little through the largely Silurian limestone stratigraphy. The extremely high within and between site directional precision of component 'I' is notable.

Red and buff sandstone

The Devonian red sediments (sites 2–13, 25, 27, 28) carry three components of NRM. The two of lowest thermal and

Table 2. Remanence components with low stability ('L').

Site N°	Age/Epoch	T_b-range	N/N_o	D/I	k	α_{95}
Dolina						
28	Gedinnian/	<200	No data			
27	Siegenian		3/6	250.3/29.3	264.6	7.6
26			Not statistically significant			
25			Not statistically significant			
Nezwiska						
9	Gedinnian/	<200	4/5	298.4/76.4	13.5	25.9
10	Siegenian		3/6	39.5/70.6	46.9	18.2
11			3/6	331.9/70.5	28.3	23.6
12			Not statistically significant			
13			No data			
Gorodnitsa						
8	Gedinnian	<250	No data			
7			Not statistically significant			
6			Not statistically significant			
5			2/6	208.7/60.4	399.8	12.5 (?)
4			Not statistically significant			
3			No data			
Zaleshchiki						
2	Gedinnian/	<200	No data			
1	Pridoli?		Not statistically significant			
1-13, 25-28	*Early Devonian*	*Sample*	*15/117*	*288.4/71.1*	*7.5*	*15.0*
		ALL	*30/117*	*286.5/72.3*	*6.6*	*11.1*
		Site	*5/17*	*287.5/70.1*	*7.8*	*29.3*
Rykhotin						
24	Pridoli	<250	9/10	18.9/82.7	28.4	9.8
23			No data			
Rashkow						
22	Pridoli/	<250	No data			
21	Ludlow?		Not statistically significant			
20			7/9	287.4/82.4	18.3	14.5
19			5/5	57.5/84.5	35.2	13.1
18			5/5	344.4/61.9	24.3	15.8
Ataky						
14	Ludlow	<300	Not statistically significant			
15			10/15	1.1/74.5	23.6	10.1
16			4/8	52.6/79.2	17.1	22.9
17			5/6	19.5/82.1	17.9	18.6
14-24	*Late Silurian*	*Sample*	*45/72*	*1.1/80.1*	*19.9*	*4.9*
		ALL	*49/72*	*355.7/81.4*	*17.0*	*5.1*
		Site	*7/11*	*3.4/80.2*	*59.0*	*7.9*
1-28	*Siluro-Devonian*	*Sample*	*60/189*	*335.9/80.1*	*12.8*	*5.3*
		ALL	*79/189*	*318.9/80.2*	*9.8*	*5.4*
		Site	*12/28*	*319.6/79.0*	*14.1*	*12.0*

Explanation for Table 2.

Sites are listed in stratigraphic order.

T_b-range=Approximate laboratory unblocking temperature range (degrees Celsius).

N/N_o=Number of samples which yielded the L-component (N)/were demagnetized (N_o).

D/I=Declination/Inclination of Fisher (1953) mean (degrees).

k=Estimate of Fisher's (1953) precision parameter.

α_{95}=Half-angle of cone of 95% confidence (degrees).

'Not statistically significant' is written where site level α_{95} exceeds 30 degrees.

Overall Fisher mean results are given at (a) 'Sample' level (including data from those sites which yielded a 'statistically significant' mean direction), (b) 'Site' level and (c) 'ALL' level (including all sample level data, even those from sites which did not yield a 'statistically significant' mean direction).

Table 3. Remanence components with intermediate stability ('I').

Site N°	Age/Epoch	T_b-range	N/N_o	D/I	k	α_{95}	I_{NRM}	$I_{Component}$
Dolina								
28	Gedinnian/	200-570	5/7	215.3/-29.5	80.4	8.6	0.2-0.4	0.1-0.2
27	Siegenian	200-610	6/6	206.1/-13.0	161.4	5.3	4.0-8.5	2.0-4.5
26		No data						
25		300-630	10/11	204.4/-12.7	236.5	3.2	2.0-6.0	1.5-6.0
Nezwiska								
9	Gedinnian/	200-490	4/5	200.3/-12.8	88.2	9.8	0.5-1.0	0.5
10	Siegenian	300-640	6/6	202.9/-15.5	81.0	7.5	2.0	1.5-2.0
11		300-630	5/6	205.3/-13.8	34.8	13.2	1.5-4.0	1.5-3.0
12		200-600	3/6	194.7/-11.2	29.6	23.1	2.0-2.5	
13		200-600	3/3	205.4/-13.2	355.6	6.6	1.5-2.0	1.5-2.0
Gorodnitsa								
8	Gedinnian	200-575	10/10	206.7/-18.6	108.6	4.7	1.5-2.0	1.0-1.5
7		200-550	9/9	204.2/-19.6	156.0	4.1	0.5-1.5	0.5-1.0
6		200-560	5/6	210.4/-17.3	899.3	2.5	1.5-3.0	0.5-2.0
5		200-580	6/6	205.1/-15.5	40.5	10.7	4.0-6.0	3.0-5.0
4		200-580	8/8	210.5/-21.2	258.5	3.5	3.0-6.0	2.5-4.0
3		150-600	7/7	212.1/-19.7	189.3	4.4	3.0-9.0	1.0-8.0
Zaleshchiki								
2	Gedinnian/	200-590	14/14	201.1/-21.2	136.2	3.4	6.0-10.0	6.0-10.0
1	Pridoli?	150-400	5/7	207.1/-15.7	45.6	11.5	0.1	0.01-0.1
1-13, 25-28	*Early Devonian*	*Sample*	*106/117*	*205.7/-17.6*	*72.7*	*4.5*		
		Site	*16/17*	*205.6/-19.9*	*151.0*	*3.1*		
Rykhotin								
24	Pridoli	270-400	9/10	207.4/-30.9	1072.3	1.6	2.5-4.5	2.5-3.0
23		200-500	8/8	205.8/-29.5	250.2	3.5	0.8-2.0	1.0-1.5
Rashkow								
22	Pridoli/	No data						
21	Ludlow?	200-500	6/6	202.0/-31.7	74.1	7.8	0.5-1.0	0.5-0.7
20		200-510	9/9	207.3/-28.5	1173.6	1.5	1.0-2.5	1.0-2.5
19		230-500	5/5	207.2/-28.0	299.9	4.4	0.5-2.0	0.5-1.5
18		200-500	5/5	202.3/-28.8	67.8	9.4	0.2-0.5	0.2-0.5
Ataky								
14	Ludlow	No data						
15		300-470	14/15	209.3/-20.0	44.0	6.1	0.5-1.0	0.1-1.0
16		250-470	4/8	205.1/-04.2	13.6	25.0	0.1-0.5	0.1-0.3
17		200-500	6/6	206.9/-16.7	390.7	3.4	3.0-5.0	4.0-4.5
14-24	*Late Silurian*	*Sample*	*66/72*	*206.5/-25.0*	*51.5*	*2.5*		
		Site	*9/11*	*205.9/-24.3*	*76.0*	*5.9*		
1-28	Siluro-Devonian	Sample	172/189	206.0/-20.4	56.2	1.5		
		Site	25/28	205.7/-19.6	95.3	3.0		

POLE: From Siluro-Devonian *Site Level*. Site Latitude=48.7°N, Longitude=26°E
Pole Latitude=45.8°S, Longitude=348.2°E, dp/dm=1.6°/3.1°, N=25 sites, Polarity=reversed.

Explanation for Table 3.

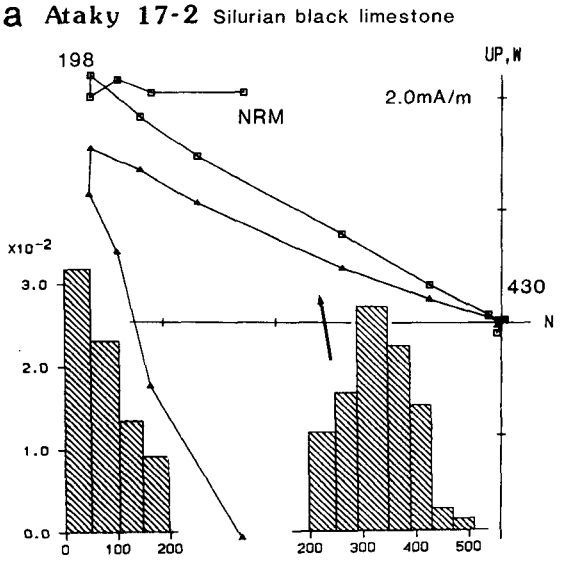
Sites are listed in stratigraphic order. Conventions are as in Table 2 except

$I_{NRM}/I_{Component}$=Approximate NRM/I-Component intensity (mA/m).

coercive force stability are reminiscent of components 'L' and 'I' carried by the older limestones, in terms of both their stability and direction (compare Figs 5–7). For this reason they too have been labelled 'L' (<300 °C, <10 mT, <10 per cent of total remanence; Table 2) and 'I' (100–630 °C, 10–>90 mT, 40–90 per cent of total remanence; Table 3).

Site level statistics for all 'L' and 'I' components are given in Tables 2 and 3, and site level 'L' and 'I' directions are illustrated in Fig. 8. Clearly the 'group of sites level' component 'I' direction for the Devonian redbeds cannot be distinguished statistically from that for the Silurian limestones (Table 3). The 'L' component constitutes a somewhat smaller proportion of total remanence in the redbeds compared with the limestones, and its direction is not precisely determined. It seems probable, however, that the underlying direction of 'L' in the redbeds is similar to that of 'L' in the limestones. Therefore *all* 'L' and *all* 'I' component directions are regarded as records of two distinct palaeofield directions (see the overall 'L' and 'I' directions, Tables 2 and 3, bottom).

Samples from most stratigraphic horizons in the Devonian sequence carry a third component of NRM with a stability spectrum higher than that of the limestone NRM. The natural (H)igh-stability component direction ('H': 570–690 °C, >>90 mT, 50 per cent of total remanence) is, however, seldom precisely defined (Table 4). Our limited success in the recovery of the 'H' component, compared, say, with the 'I' component, can be attributed to the narrow



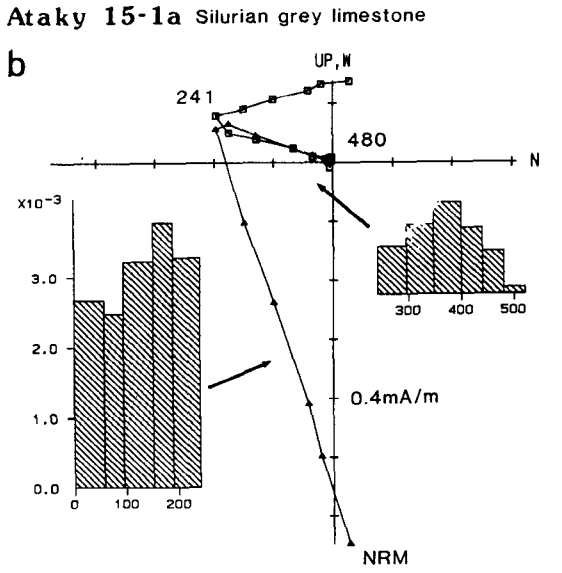


Figure 5. Orthogonal plots of *in situ* demagnetization trajectories: Silurian limestones. Note that squares (resp. triangles) lie in the horizontal (resp. vertical) plane. Treatments are in degrees Celsius. Unblocking temperature spectra: *X* axes—degrees Celsius applied heat; *Y* axes—mA m^{-1} (remanent magnetization removed) per degree Celsius applied heat. The area of blocks represents the remanent magnetization removed in one treatment interval.

unblocking temperature spectrum of 'H', which frequently overlaps with that of 'I'. Furthermore, the exact nature of 'H' is occasionally obscured by 'noisy' demagnetization behaviour caused by high-temperature mineralogical alteration in the furnace. Nevertheless, the direction of 'H' has been clearly isolated at sites 2, 4, 6–11, 25, 27 and 28; 11 (Devonian) sites out of a total of 15. One of the most significant features of these data is that 'H' is of dual polarity [Fig. 9; compare Fig. 6(b–e) with Figs 7(b) and 11(e); Table 4]. The integrity and wide stratigraphic extent of 'H' is confirmed by numerous curved demagnetization trajectories (e.g. Fig. 6e).

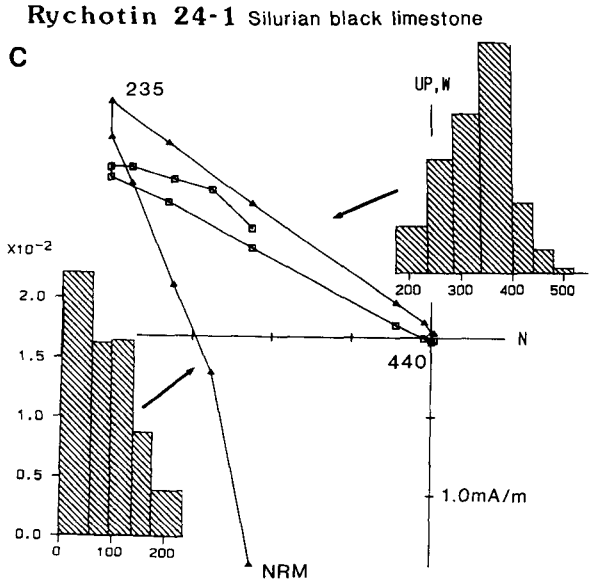


Figure 5. (*continued*)

Table 4. Remanence components with high stability ('H').

Explanation for Table 4.

Sites are listed in stratigraphic order. Conventions are as in Table 2 except 'Sample'=sample number, 'Polarity' N=normal R=reversed.

Site N°	Age/Epoch	T_b-range	N/N_o	D/I	Sample	Polarity	
Dolina							
28	Gedinnian/ Siegenian	570-690	5/7	254.2/32.2	(1)	R	
				235.5/31.2	(2)	R	
				235.3/22.3	(3a)	R	
				48.7/-15.2	(4)	N?	
				257.7/40.8	(5)	R	
27	*Gedinnian/ Siegenian*	*570-690*	*5/6*	*194.2/16.7*	*(1)*	*R?*	
				237.0/27.0	*(2)*	*R?*	
				259.0/4.9	*(3)*	*R?*	
				235.1/14.9	*(4)*	*R?*	
				211.8/10.7	*(5)*	*R?*	
25	*Gedinnian/ Siegenian*	*570-690*	*8/11*	*217.8/14.0*	*(1b)*	*R?*	*Not included in mean calculation*
				185.4/15.0	*(2a)*	*R?*	
				190.7/14.1	*(2b)*	*R?*	
				194.1/25.9	*(3)*	*R?*	
				171.0/25.0	*(4)*	*R?*	
				124.1/29.6	*(6)*	*R?*	
				214.8/-6.0	*(7a)*	*R?*	
				204.9/10.0	*(7b)*	*R?*	
Nezwiska							
9	Gedinnian/ Siegenian	570-690	2/5	265.8/53.0	(3b)	R	
				236.0/18.2	(4)	R	
10		570-690	1/6	255.0/30.7	(1)	R	
11		570-690	2/6	106.8/-14.9	(1)	N	
				45.9/-32.5	(3)	N	
Gorodnitsa							
8	Gedinnian	570-690	1/10	65.2/-49.7	(1a)	N	
7		570-690	3/9	45.9/-27.6	(1b)	N	
				90.0/-38.2	(4c)	N	
				61.0/-32.4	(5a)	N	
6	Gedinnian	570-690	4/6	75.1/-28.0	(3)	N	
				81.7/-26.4	(4)	N	
				72.8/-34.2	(6a)	N	
				63.4/-35.1	(6b)	N	
5 'H' not well defined by data: polarity seems to be NORMAL						N?	
4	Gedinnian	570-690	6/8	84.1/-67.0	(1)	N	
				87.0/-51.4	(2b)	N	
				104.0/-70.5	(3b)	N	
				68.0/-61.1	(5a)	N	
				70.0/-50.2	(6a)	N	
				75.1/-54.2	(7)	N	
3 'H' not well defined by data: polarity seems to be reversed						R?	
Zaleshchiki							
2	Gedinnian/ Pridoli?	570-690	1/14	235.0/28.4	(1-4a)	R	
2-11, 28	Early Devonian		Sample	25/100	249.4/38.8, k=16.6, α_{95}=7.3		

POLE: From Devonian *Sample Level*. Site Latitude=48.7°N, Longitude=26°E
Pole Latitude=3.7°N, Longitude=325.5°E, dp/dm=5.2°/8.7°, N=25 samples, Polarity=mixed.

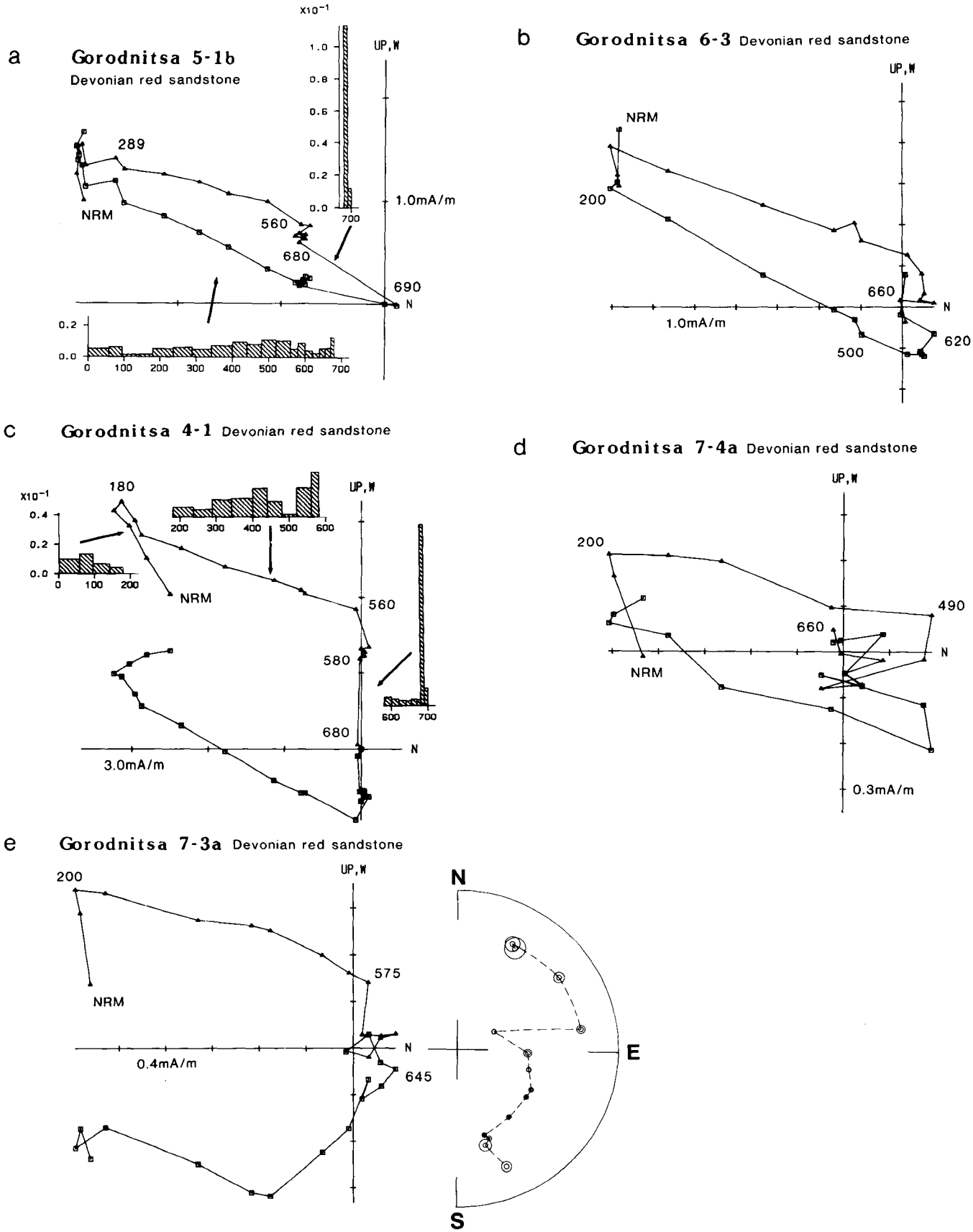


Figure 6. Orthogonal plots of *in situ* demagnetization trajectories: Devonian redbeds. Illustrative conventions are as in Fig. 5 except (e): equal angle projection of demagnetization data including 95 per cent measurement error estimates. The data from demagnetization temperatures exceeding 645 °C (7-3a) are regarded as unreliable.

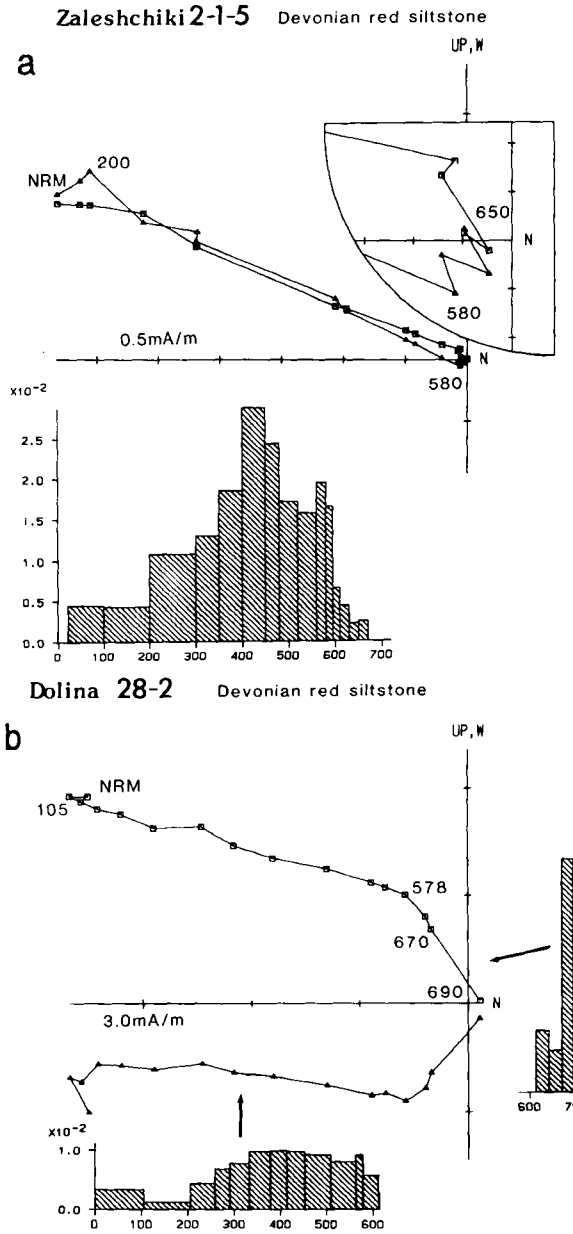


Figure 7. Orthogonal plots of *in situ* demagnetization trajectories: Devonian red siltstones. Illustrative conventions are as in Fig. 5 except enlargements of the trajectories in (a) and (c) are given as insets.

Figure 10 illustrates the dual polarity 'H' component directions of Fig. 9 along with high-stability remanence directions extracted from demagnetization trajectories like the one in Fig. 6(a). Those high-stability 'components' are not punctuated by data and might simply be vector sums of the 'H' (normal polarity) and 'I' components. If this is true, then we can amend Table 4 to include sites 5 and 8 as recording the normal polarity 'H' component. If not, we might regard the directions intermediate to 'I' and 'H' as recent reversed polarity overprints.

The 'H' remanence components from sites 25 and 27 have much in common with the others, but offer supplementary information regarding the mode of origin of 'H'. A later,

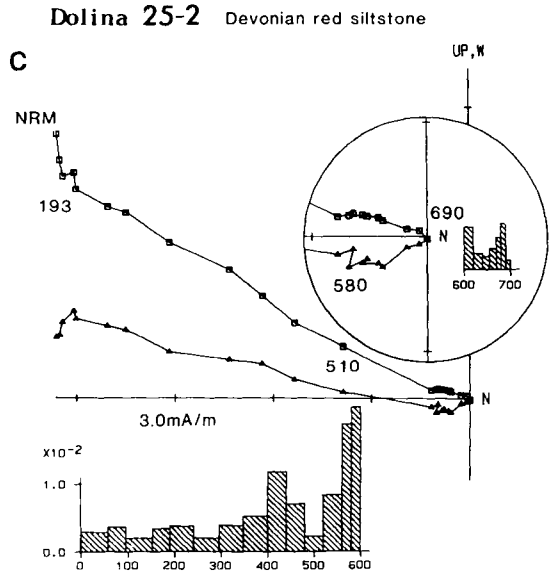


Figure 7. (*continued*)

crucial, section of this paper is dedicated to the description of data from those two sites.

COMPONENTS OF REMANENCE FROM ASSOCIATED ROCKS

It is significant to note at this point that many Mid- and Lower Palaeozoic rock units, situated throughout the

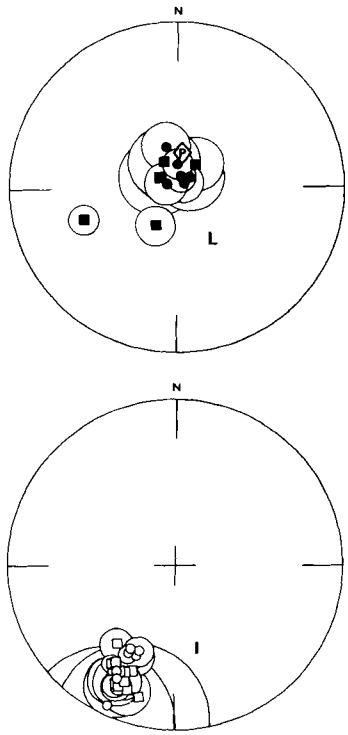


Figure 8. Equal angle projection of 'L' component (TOP) and 'I' component (BOTTOM) site mean directions with 95 per cent confidence limits. Open (resp. closed) symbols lie on the upper (resp. lower) hemisphere. Squares (resp. circles) denote data from Devonian (resp. Silurian) rocks. The present-day Earth's field direction in Podolia is indicated ('P', TOP).

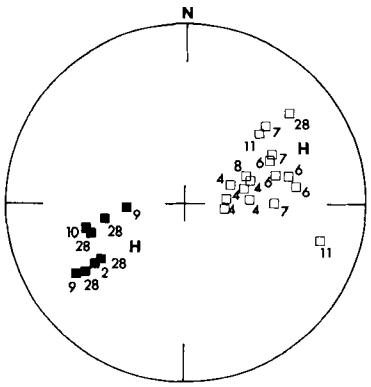


Figure 9. Equal angle projection of sample level 'H' component directions from the Devonian rocks. Site numbers are indicated. Illustrative conventions are as in Fig. 8.

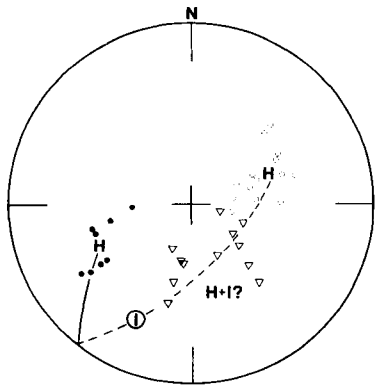


Figure 10. Equal angle projection of the 'H' component data of Fig. 9 (small circles) and poorly defined high-stability directions (triangles) at sample level. 'I' denotes the overall mean (upward pointing) direction of the 'I' component. Note that the triangles fall, approximately, on a great circle path between 'I' and the normal polarity 'H' directions. Illustrative conventions are as in Fig. 8.

Russian platform, carry a remanence component correlative with 'I' of this study [Fig. 11; (b) Silurian limestone from lower in the Dniester River section; (c) Cambrian red siltstone from western Ukraine; (d) limestone from the Main Devonian Field near Leningrad].

It is also relevant to note that a set of 20 Devonian samples from Podolia, analysed by MAS while at Leeds University (Fig. 11e), clearly show the reversed polarity 'H' component trend described above (Fig. 9).

MAGNETIC FABRIC

The anisotropy of low-field susceptibility (AMS) of all samples was measured. A weak flat-lying magnetic foliation was discernable in samples from all sites, corresponding to the primary bedding fabric S_0 (also flat lying). In addition, samples from fine-grained red siltstones at sites 25 and 27 were found to possses a magnetic lineation. That lineation is the subject of the following discussion.

Sites 25 and 27 are located at Dolina in the uppermost red siltstones in the sampling traverse. Site 25 lies in red siltstone overlain by a horizontal grey sandstone unit (site 26). Contorted laminae in the grey sandstone unit define ancient slump structures. Sites 27 and 28 lie in the overlying flat-bedded red siltstones.

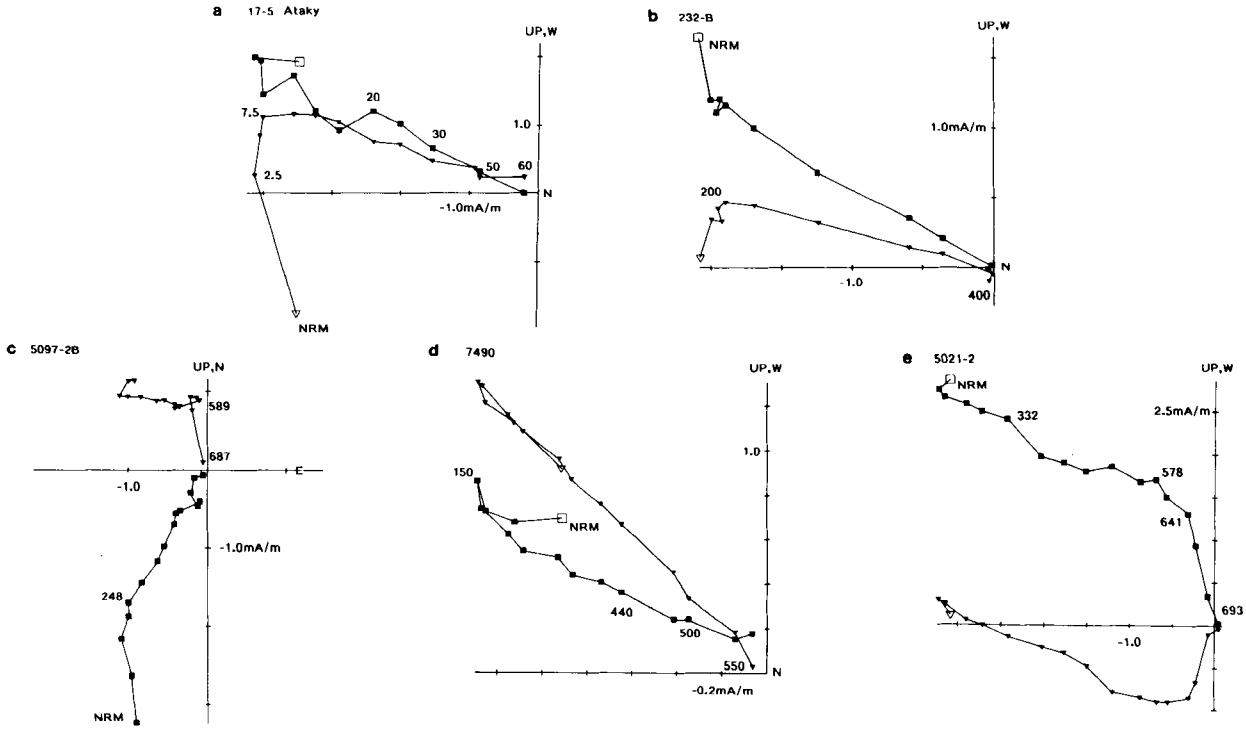


Figure 11. Orthogonal plots of *in situ* demagnetization trajectories: (a) alternating field demagnetization of Silurian limestone (treatments in mT); (b–e) results from earlier made sample collections, demagnetized by MAS and ANK; (b) Silurian (S_2) limestone, Podolia, Slobodka Ustievo horizon: (c) Cambrian red siltstone, Ukraine; (d) Devonian (D_3) limestone, Main Devonian Field, river Kolozhka; (e) Devonian (D_1) red siltstone, Podolia, Dniester River section. Note that fold tests are not available for the results illustrated in this figure.

Remanence components 'L', 'I', and 'H' were identified in a number of samples from sites 25 and 27. The 'L' and 'I' components resemble their namesakes from elsewhere in the stratigraphy in terms of stability, direction and dispersion. Similarly, 'H' has its characteristic stability and reversed polarity. Uncharacteristically, 'H' at 25 and 27 is widely dispersed in declination between south and west. The samples yielding those dispersed directions have a weak magnetic lineation superimposed on the usual bedding parallel foliation (maximum:intermediate = 1.01 to 1.04). Comparison of remanence directions and the orientation of corresponding AMS ellipsoids reveals a positive correlation between the two.

To illustrate this correlation the dispersed 'H' component declinations have been rotated about a vertical axis (the minimum susceptibility axis) by an amount which brings the maximum susceptibility axes into alignment with an arbitrary reference direction (Fig. 12). This results in a marked improvement in the declinational distribution of the remanence vectors. Any interpretation of this phenomenon must also take into account the fact that the aforementioned 'H' component declinations tend to bisect the angle between maximum susceptibility axes and the west–southwest 'H' component trend from elsewhere in the stratigraphy, where no clear remanence-fabric pattern was recognized.

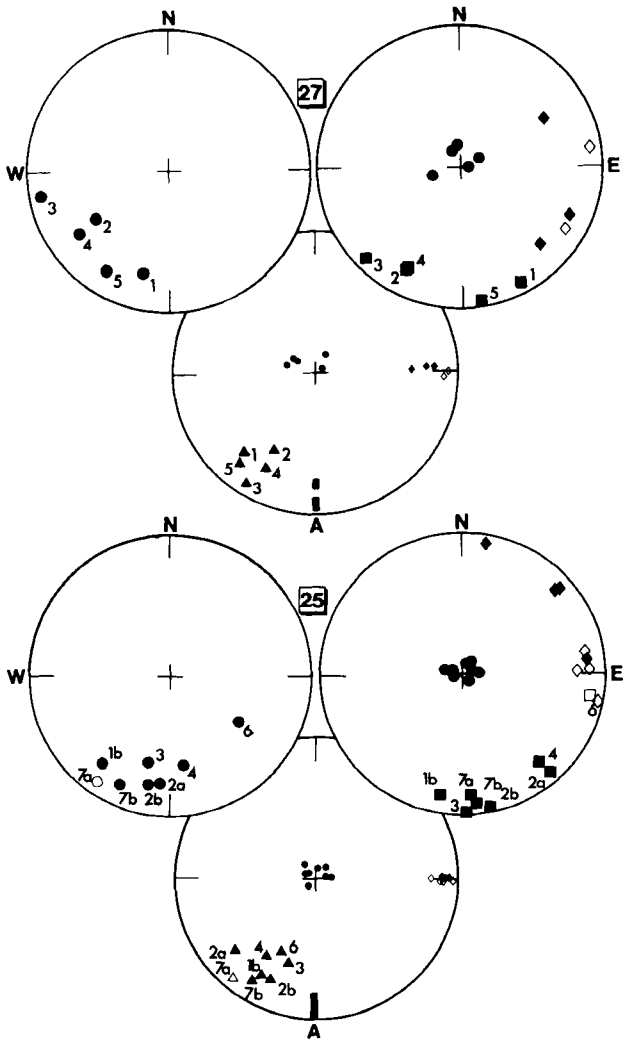


Figure 12. 'H' component remanence directions and anisotropy of magnetic susceptibility at sites 25 and 27. The top three equal angle projections display data from site 27. The bottom three display data from site 25. The following explanation pertains to both groups of three projections. Upper left: *in situ* 'H' component directions; upper right: anisotropy of magnetic susceptibility (maximum axes = squares, intermediate = diamonds, minimum = circles); Lower centre: the anisotropy data (small symbols) with maximum axes translated to the arbitrary reference direction 'A' (about a vertical axis), triangles = remanence directions translated by the same amount. Sample numbers are indicated beside *in situ* remanence directions and maximum susceptibility axes. Remanence directions are listed in Table 4. Illustrative conventions are as in Fig. 8.

MINERALOGY

Haematite

The 'H' component, observed in the redbeds, is stable between the Curie points of magnetite and haematite (Figs 6, 7, 11), and has a coercive force stability spectrum exceeding 100 mT. The coercivity of remanence of samples carrying 'H' is greater than 80 mT, and isothermal remanence saturation is not achieved with inducing fields up to 1.2 T, the maximum direct field available. This clearly indicates a haematite host for the 'H' component. Haematite is evidenced in polished rock thin section in specular and pigment form. It is likely that much of both forms are early (Early Devonian) features because (1) at least some of the specular haematite appears to be detrital and (2) the reddening caused by the haematite pigment is stratigraphically controlled.

Magnetite

The 'I' component is stable between 100° and 630 °C (10–>90 mT) in the redbeds and 200°–510 °C (7–70 mT) in the limestones. With regard to the limestones one might expect a magnetite host, which indeed is evidenced in polished rock thin section, along with pyrite in the darker coloured varieties. We therefore presume a magnetite host for 'I' in the limestones. Grains of magnetite were also identified in the redbeds, however the upper thermal stability limit of 'I' in the redbeds (630 °C) exceeds the Curie point of magnetite. We therefore conclude that 'I' in the redbeds might be carried by both magnetite and the more stable haematite, and therefore that there might be two *genera* of haematite.

INTERPRETATION

The 'L' component

The 'L' component, observed at most sites but best defined in the Silurian limestones has a low stability and direction approximately parallel to the present-day Earth's field direction in Podolia ($D = 5°$, $I = 66°$). Therefore we presume that 'L' is a viscous remanent magnetization of comparatively young age and limit further discussions to the 'I' and 'H' components.

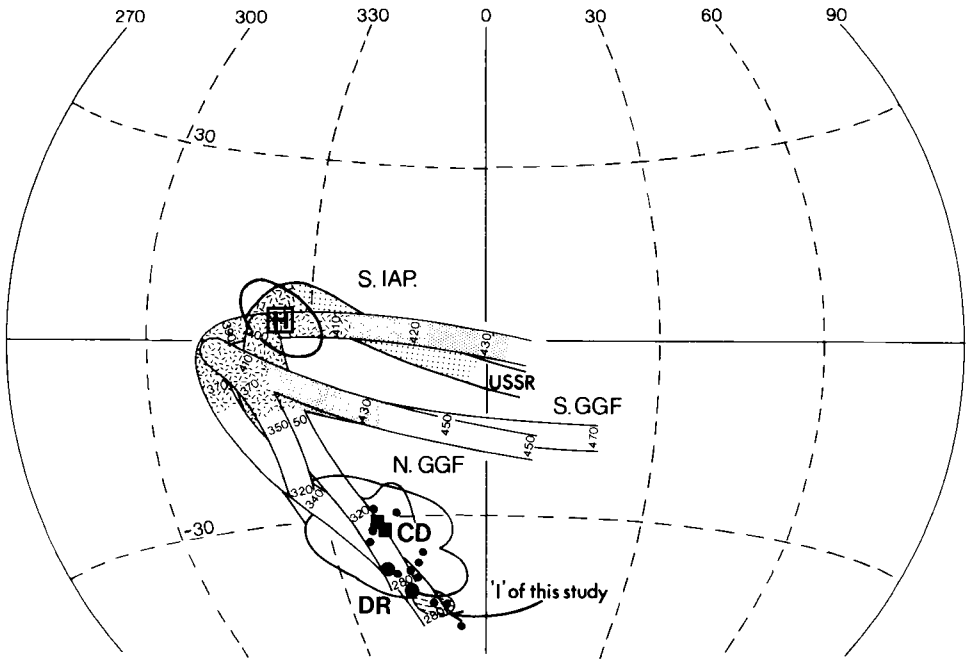


Figure 13. Reference APW paths for Britain and the Russian plate from Figs 1 and 3 along with the mean 'H' component pole position ('H') and associated error oval. The 'H' component data are listed in Table 4. The 'I' component pole is plotted with horizontal dashed error oval (listed in Table 3).

The 'I' component

It has been demonstrated that the exceedingly well-defined 'I' component can be isolated throughout the Silurian and Devonian sequences sampled as part of this study. Equivalent remanence components, with similar directions and stabilities, are carried by Mid-Palaeozoic sediments outcropping elsewhere on the Russian platform and by Lower Palaeozoic sedimentary rocks in western Ukraine (Figs 11 and 13; Hicken *et al.* 1972; Khramov *et al.* 1981; Khramov 1984, 1986).

The palaeomagnetic pole position corresponding to 'I' (overall pole, Table 3) falls within the group of Russian poles (of uncertain age) which lie on the Upper Palaeozoic part of the British reference path (Fig. 1) and our Russian APW model (Fig. 13). Those poles are from Mid-Palaeozoic sediments of the Russian platform, including the successions sampled in this study. We draw two conclusions from this. The first is that our 'I' component corresponds in direction to the magnetizations reported by others to be of primary origin. The second is that the 'I' component and those earlier reported magnetizations are most likely to be of secondary Upper Palaeozoic (Permian?) age.

The 'H' component

Dual polarity

The occurrence of both *normal* and *reversed* polarity 'H' component directions in the Devonian sediments suggests a primary, Early Devonian remanence age. Normal and reversed polarities are with respect to the Devonian part of the reference south APW paths for Britain (Fig. 1), and our model Russian APW path (Fig. 3). By means of a summary, the lower part of the Devonian sequence at Zaleshchiki and Gorodnitsa records a reversed polarity palaeofield (Gedinnian/Pridoli, sites 2 and 3). The rest of the Gorodnitsa profile, and the Nezwiska sequence indicate normal polarity (Gedinnian–Siegenian, sites 4–11). The Dolina sediments indicate a return to a reversed polarity Devonian palaeofield (Siegenian/Gedinnian, sites 25–28; Table 4).

Magnetic mineralogy

Optical mineralogical and field observations suggest an early origin for much of the haematite in the red beds. We are confident from palaeomagnetic laboratory experiments that haematite is the host mineral phase for 'H' and therefore conclude that our magnetic mineralogical investigations support a primary or near-primary origin for the 'H' component.

Magnetic fabric

The ordered geometrical relationship between the 'H' component directions and magnetic fabric (AMS) at sites 25 and 27 indicates some causal relationship between the two. Therefore we attempt to formulate a model which embraces (1) low-field anisotropy, (2) the haematite remanence ('H') and (3) sedimentary/diagenetic processes.

We assume that the measured low-field anisotropy is associated with the magnetocrystalline anisotropy of haematite. In the case of platy haematite we might be justified in assuming that the grain shape anisotropy is broadly equatable (concordant) with the magnetocrystalline anisotropy. From this we deduce that the measured anisotropy reflects a weak haematite grain alignment,

introduced through the action of sedimentation and/or diagenetic processes.

We observed a fairly consistent angular relationship between maximum susceptibility axes (K_{max}) and remanence vectors, where remanence directions tended to bisect the angle between K_{max} and what we presume to be the regional palaeofield direction (west–southwest and down: Fig 9). That geometry bears a striking resemblance to patterns of behaviour observed by King (1955) during laboratory sedimentation experiments, in which the action of currents caused the remanence direction of accumulating sediment to form oblique to the ambient field direction. Whatever the exact explanation of our observations on ancient rocks, those observations strongly support a detrital remanent magnetization (DRM) mode of origin for 'H' at sites 25 and 27. In so doing they support our preceding arguments for an Early Devonian remanence age.

Pole position

Although the pole positions corresponding to sample level 'H' component directions are few and scattered, they agree reasonably well with the Devonian part of the Bristish reference paths and with our path for the Russian plate, of which the Mid-Palaeozoic part is interpolated (Fig. 13). Given the objective of this investigation this line of argument can only offer circumstantial evidence for remanence age.

CONCLUSIONS

A Late Palaeozoic (or Early Mesozoic) pervasive remagnetization event affected Palaeozoic sedimentary rocks of the Russian platform. It seems likely that many of the published palaeomagnetic results from Mid-Palaeozoic rocks of the Russian platform can be attributed to this event.

The Devonian and Silurian rocks outcropping on the banks of the Dniester River, western Ukraine, carry a secondary magnetization component, probably of Late Palaeozoic/Early Mesozoic age, attributable to the aforementioned pervasive remagnetization event (latitude = 45.8°S, longitude = 348.2°E, $dp/dm = 1.6°/3.1°$, $N = 25$ sites, reversed polarity; Table 3).

The Devonian sediments in western Ukraine carry a high-stability remanence component of probable primary Devonian age. The age determination of this magnetization is founded on (1) its dual polarity, (2) strong evidence for a DRM mode at two localities, (3) a detrital haematite and/or early diagenetic haematite host and (4) its directional consistency with Devonian data from Britain. [Note that condition (4) depends on the 'Old Red Continent' being assembled in Devonian time.] The direction of this component was established at a number of sites and provides a good indication of the Devonian pole position for the Russian plate and Baltica (latitude = 3.7°N, longitude = 325.5°E, $dp/dm = 5.2°/8.7°$, $N = 25$ samples, mixed polarity; Table 4). This is the only palaeomagnetic result from well-dated Devonian rocks of Baltica (Scadinavia and the Russian plate) which (1) can be demonstrated to be of primary Devonian age and (2) is not invalidated for palaeogeographic analysis by regional tectonic complexities.

General agreement between the Devonian pole position for the Russian plate as indicated by the high-stability remanence component of this study and the Devonian part of the APW path for Britain places the two continents in their present relative position in Devonian time.

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